

Gesture Inference on the Arduino Nicla Vision

AI Driven Embedded Systems

Hardware Insights: Arduino Pro Nicla Vision

- **Required Quantization to be switched from Int8 to Float.**
 - For unknown, undocumented reasons, the nicla will refuse to allocate memory for a model quantized into anything but Float32.
- Required me to downsize image resolution from 64x64 to 32x32 to prevent tensor allocation failure*

*I actually don't think that's necessary, as the real culprit of the allocation failure was the int8 quantization



Dataset Properties

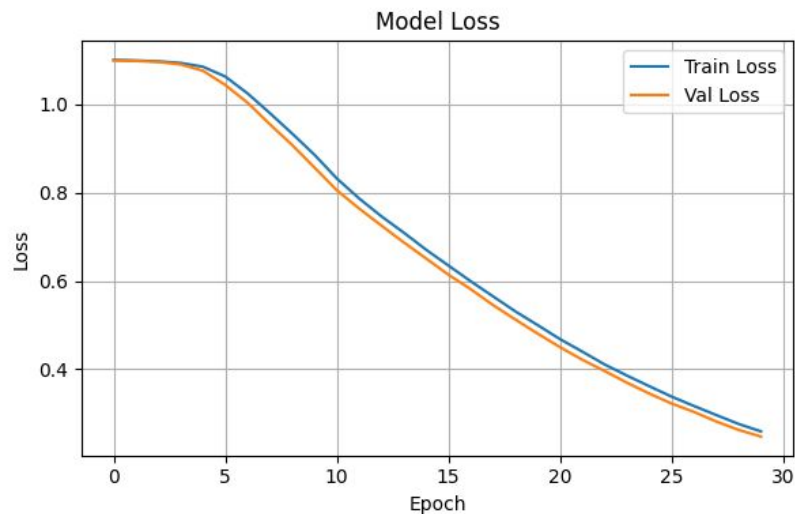
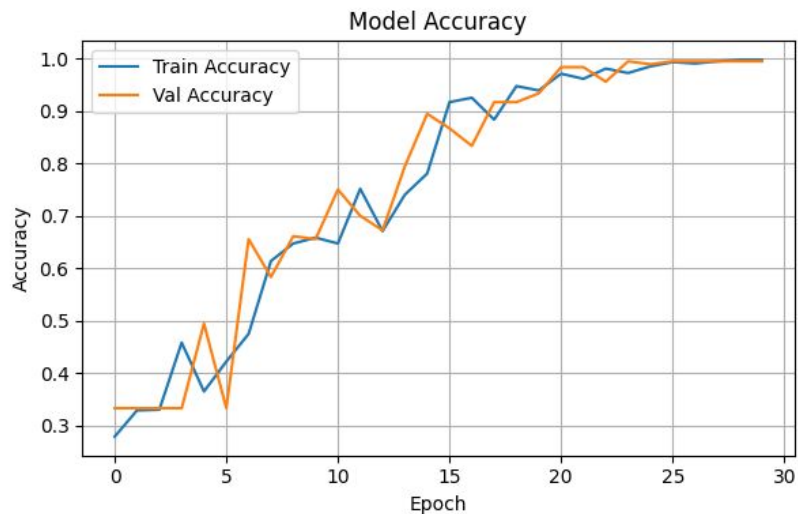
1. 32x32 grayscale .pgm image

- a. 32x32 was chosen to keep resulting .tflite model small to fit within the nicla's brutal memory constraints.
- b. The nicla actually does not have a native 32x32 snapshot feature, so I actually captured 64x64 images and shrunk them down for the dataset.
 - i. I repeated that process for capturing images in the inference script.
- c. 32x32 is possibly unnecessary, as I discovered the true culprit of tensor allocation failure is that the nicla can't accept models with int8 quantization, only float.

2. 300 images per label (LEFT, RIGHT, NONE), captured against black background...

- a. ...in my room, around 3:30 pm on a sunny day.

Training Visualization



Resulting Model Properties

1. Size: 3.8 Kb
2. Functions ok under very specific circumstances, definitely needs more data.
Was able to correctly recognize gestures but only at certain time of day, in a certain location.

Reflection

1. Final model size (KB)?

3.5 kb

2. Quantization impact on accuracy?

Seemingly none, judging by the fact that the TFLite Model and Keras model yield 100% accuracy.

3. Why is this model suitable for OpenMV?

It's very small, it uses GlobalAveragePooling2D instead of large fully connected layers, it operates on low resolution 32x32 grayscale images, and the inference happens entirely on the microcontroller requiring no internet connection.

4. What would change for a different sensor?

Our input pipeline would need to change to match the new sensors output format. For example, if the sensor outputs color (RGB) images, the number of input channels would increase from 1 to 3, requiring changes to the input_shape, the dataset preprocessing, and likely the model capacity (more filters). If the resolution differed, the resize() step would need to be updated. The data capture script would also need to be rewritten to use the new sensor's API.